

Effect of Cultivation on the Relationship between Root Length Density and Unsaturated Hydraulic Conductivity in a Moderately Swelling Clay Soil

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Abstract

Few studies have examined the effect of cultivation on the relationship between the presence of roots and unsaturated hydraulic conductivity in the soil matrix. Roots change soil pore characteristics by modifying pore diameter and pore continuity. Thus, the presence of roots may be expected to affect the hydraulic conductivity in saturated and unsaturated phases.

In situ measurements of unsaturated hydraulic conductivity (K_u) at 0.04 m suction and root length density (L_v) of a moderately swelling soil were made in a wheat field and in a natural pasture field at 0, 0.25 and 0.4 m depths to determine their inter-relationships. The K_u value decreased significantly with depth and the surface layer of the wheat field had the highest value (0.28 m day⁻¹). The pasture field had high L_v at the surface layer (7.3 cm cm⁻³), but values decreased to less than those in the wheat field at 0.4 m depth. There was a distinct positive correlation between L_v and K_u , with the slope of the linear regression line being greater in the wheat field than in the pasture field. The cause of the relation is probably due to the common pore characteristics which influence K_u and L_v .

Introduction

Apart from brief periods following rainfall or irrigation and subsequent drainage of free water, transport of soil water within the root zone will occur in the unsaturated phase. The extent of such unsaturated flow is determined by unsaturated hydraulic conductivity, K_u (m day⁻¹), and depends on the size, distribution and continuity of soil pores. Soil water flow and root growth generally follow the path of least resistance within the soil matrix, which is offered by continuous soil pores. Hamblin and Tennant (1981) found that a reduced portion of large pores on a zero-tilled soil resulted in lower infiltration rates and reduced root growth. This situation persisted unchanged over a 5 year continuous cropping period (Hamblin *et al.* 1982).

In the situation where roots grow into and thus fill a high proportion of soil pores, it might be expected that K_u will decrease as root length density, L_v (cm cm⁻³), increases. Barley (1953) reported that root growth resulted in a reduction in hydraulic conductivity at saturation. Sedgley and Barley (1958) found that unsaturated hydraulic conductivity at 0.3 m suction was also reduced by root growth. Both observations were explained in terms of a reduction in pore space, as a result of blocking of pores and compaction of the surrounding soil matrix due to radial growth of individual roots. Gish and

Jury (1983) reported that roots increased the amount of immobile water within the matrix, even after root death, resulting in lower hydraulic conductivity.

The experiments described above were short-term, conducted in repacked sandy loam columns where the soil structure is relatively weak. Hence, reduction in porosity and pore diameter could occur through blocking of pores and some rearrangement of soil particles by root growth. This is not necessarily true in fine textured, moderately swelling clay soils, where root growth could alter the pore geometry, create relatively larger continuous pores and reduce tortuosity without altering the total porosity over a period of time. Further, shrinkage around roots due to water uptake could result in finer 'conduits' for unsaturated flow of water.

Roots are known to create continuous inter-connected permanent channels (Ehlers *et al.* 1983) and improve the stability of soil structure by producing organic substances capable of binding the soil particles (Tisdall and Oades 1982). The average diameter of first- and second-order lateral roots of cereals ranges from 20 to 50 μm (Nye and Tinker 1977), whereas the diameter of pores involved in redistribution of water in the unsaturated phase ranges between 0.1 μm (matric potential, $\psi_m = -1500$ kPa) and 30 μm ($\psi_m = -10$ kPa). Such an observation suggests that the rearrangement of finer pores by first- and second-order lateral roots may play an important role in determining unsaturated flow of soil water within the root zone.

Therefore, we could anticipate that, even if active root growth reduces hydraulic conductivity in a moderately swelling soil, the decayed roots, partially decaying roots and roots which have ceased growth could increase hydraulic conductivity by improving soil structure and increasing the percentage of relatively larger pores. Evidence presented by Emerson (1954), Petersson *et al.* (1987) and Rengasamy and Mehanni (1988) supports such a premise. Emerson (1954) observed that permeability increased as roots started to die and decay in a heavy clay soil. Petersson *et al.* (1987) observed that the mass of root and saturated hydraulic conductivity (K_s) of haplic Xerosols and calcic Fluvisols in Tunisia were positively correlated. Rengasamy and Mehanni (1988) observed that the infiltration rate of a pasture field in a moderately swelling illitic clay soil increased with time, presumably as root systems interacted with the soil matrix. These studies relate infiltration rate or K_s to the actual presence of roots, assuming existing root mass is a satisfactory indicator of recent root activity within the soil matrix.

Like those studies above, most studies have been concerned with the inter-relation between K_s and L_v . However, the relation between K_u and L_v is of value in indicating more directly the extent of unsaturated flow in the critical top soil layers. Therefore, the objective of the present study was to determine the relation between K_u and L_v of a moderately swelling soil *in situ*.

Methods

Site

The study was conducted at Whitton, N.S.W., 34° 32' S., 146° 11' E. The site was mapped in the Mundiwa Association by van Dijk (1961) and the soil fits the description of Mundiwa clay loam, except that the A horizon depth of 0.18 m was greater than the more typical 0.08–0.1 m. The texture of the A horizon was a fine sandy clay loam to clay loam with 17%

clay (20% kaolinite and 50% illite) and the B horizon had 62% clay (35% kaolinite and 35% illite) (Loveday *et al.* 1984). Since the soils do not undergo drastic volume changes due to shrinking and swelling, like Vertisols, and they have higher clay content, they are described as moderately swelling clay soils. The relations between bulk density and volumetric water content of similar soils were reported by Jayawardane and Greacen (1987).

The K_u and L_v measurements were made in a wheat field a month after harvest and in an uncultivated natural pasture field at the same time. The irrigated wheat field contained only dried standing stubble, while the pasture field had annual natural pasture (mainly barley grass *Hordeum leporinum* and rye grass *Lolium species*) with 50% of the plant still actively growing. The wheat field was initially cultivated in 1983, followed by wheat crops in 1984, 1985 and 1987 and a rape (*Brassica campestris*) crop in 1986. In general, cultivation and land preparation were done when the soil was relatively moist.

Unsaturated Hydraulic Conductivity K_u

The K_u measurements at 0.045 m suction were made at 0, 0.25 and 0.4 m depths in both fields with a disc permeameter (Perroux and White 1988). The internal diameter of the water reservoir was 0.05 m and the disc was 0.2 m. Irrigation water of approximately 2 m.e. l^{-1} was used as the permeating fluid. The Na^+ , Ca^{2+} , Cl^- and SO_4^{2-} ions were the major ions present in the water used. The time required for the water reservoir level to drop 0.01 m was recorded up to a maximum of 0.2 m drop. The duration of a measurement ranged between 20 and 40 min. Ten separate measurements were made at each depth. Each measurement was separated by 0.5 m to include random variations but to minimize any spatial soil variations within the field.

Bulk Density ρ_d

The ρ_d value was determined from undisturbed core samples of $2.45 \times 10^{-5} m^3$ in volume. Samples were obtained by inserting a brass ring (0.05 m in length, 0.025 m internal diameter and 0.03 m external diameter), approximately 0.1 m from the K_u measurement locations. Samples were placed individually in air-tight containers. The initial weight (w_i) of the sample was divided by the core volume to determine ρ_d at initial water content θ_{vi} . Each sample was oven-dried overnight and re-weighed (w_d). Initial gravimetric water content θ_{gi} was equal to $(w_i - w_d)/w_d$ and θ_{vi} was equal to $\theta_{gi} \rho_d$. Subsequently, averages of ten ρ_d and θ_{vi} values for each depth were determined.

Root Length Density L_v

Ten soil cores of 0.05 m diameter and 0.15 m long were obtained at each depth at or very near the locations where K_u measurements were made. Each core was then divided into three subsamples of length 0.05 m. Samples were soaked overnight in 5% 'calgon' solution and mechanically shaken for 1 h, before the soil was manually washed away from the roots through sieves of 850 μm square mesh size. The L_v value of the sample was then determined by image analysis (Barrs 1985).

Statistical Analysis

Analyses of variance were conducted to determine whether K_u and L_v differed significantly between depths and fields. Further, mean K_u at each depth was regressed against mean L_v at corresponding depths to determine the correlation between K_u and L_v .

Results and Discussion

Unsaturated Hydraulic Conductivity K_u

Mean ($\bar{X}$) and standard deviation (s.d.) of measured K_u , mean soil bulk density (ρ_d) and mean initial volumetric water content (θ_{vi}) and the analysis of variance for K_u measurements are presented in Table 1 and Table 2. The K_u value generally decreased with depth, but only the surface K_u value in the wheat field was significantly ($\alpha=0.05$) greater than those in the deeper

Table 1. Mean unsaturated hydraulic conductivity K_u (m day⁻¹) at 0.04 m suction, mean bulk density ρ_d and mean initial volumetric water content θ_{vi} in wheat and pasture fields

Depth (m)	Wheat					Pasture				
	$\bar{X}$	K_u	s.d.	ρ_d	θ_{vi}	$\bar{X}$	K_u	s.d.	ρ_d	θ_{vi}
0	0.28 ^A	0.13		1.25	0.08	0.16	0.10		1.35	0.05
0.25	0.12	0.17		1.36	0.21	0.13	0.08		1.37	0.17
0.40	0.03	0.01		1.53	0.22	0.10	0.03		1.44	0.18

^A l.s.d._($\alpha=0.05$) = 0.09.

Table 2. Analysis of variance of unsaturated hydraulic conductivity, K_u , measurements

Source	SS	DF	MS	F	P
Depth	0.22	2	0.11	10.24	0.00
Field	0.00	1	0.00	0.36	0.55
Depth×field	0.12	2	0.06	5.67	0.01
Error	0.58	54	0.00		

Table 3. Root length density, L_v , in wheat and pasture fields (cm cm⁻³)

Depth (m)	Wheat		Pasture	
	$\bar{X}$	s.d.	$\bar{X}$	s.d.
0.00–0.05	5.1 ^A	2.0	7.3	3.7
0.05–0.10	4.9	1.3	4.7	1.6
0.10–0.15	3.1	1.3	5.0	1.6
0.25–0.30	2.3	1.3	3.9	1.6
0.30–0.35	1.5	0.8	2.4	0.8
0.35–0.40	2.1	1.2	2.6	1.0
0.40–0.45	2.0	0.8	1.9	0.6
0.45–0.50	1.8	0.3	1.3	0.4
0.50–0.55	1.7	1.2	1.2	0.5

^A l.s.d._($\alpha=0.05$) = 0.75.

Table 4. Analysis of variance of root length density, L_v , measurements

Source	SS	DF	MS	F	P
Field	18.6	1	18.6	8.6	0.00
Depth	437.0	8	54.6	25.1	0.00
Depth×field	48.9	8	6.1	2.8	0.01
Error	325.6	162	2.1		

layers. The K_u value did not differ significantly between fields when all depths were considered. However, the interaction between depth and field on K_u was significant (Table 1). This implies that the hydraulic conductivities at a particular depth between the fields differed significantly.

Cultivation in the wheat field appeared to have resulted in compaction of deeper layers (0.4 m) and subsequent reduction in K_u . The pasture field, having never been cultivated, did not show major differences in K_u with depth.

Observed differences were due to changes in soil texture from a fine sandy clay loam in the A horizon to clay loam and clay in the B horizon (Loveday *et al.* 1984).

Root Length Density L_v

The L_v values observed from the pasture and wheat fields and the analysis of variance are presented in Table 3 and Table 4. The variation in L_v between fields, depths and interaction between field and depth were significant ($\alpha = 0.05$). The mean and standard deviation of L_v of pasture field was higher than that of wheat field. Although both wheat and pasture had highest L_v in the surface layer, root distribution under wheat was more evenly distributed with depth than under pasture. This may be due to the shallow rooting nature of many pasture species and/or a result of cultivation of wheat field.

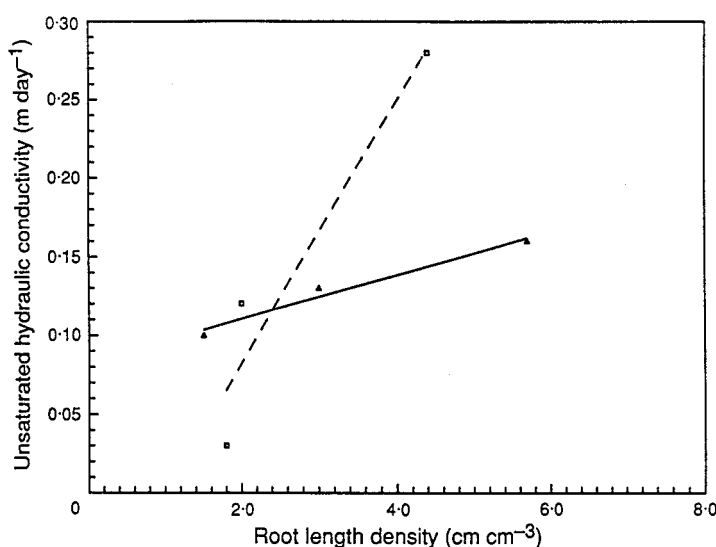


Fig. 1. Relation between mean unsaturated hydraulic conductivity K_u at 0.04 m suction and mean root length density L_v from a 0.15 m soil core ($\square$, wheat field; $\triangle$, pasture field).

Correlation between Unsaturated Hydraulic Conductivity K_u and Root Length Density L_v

Correlation coefficients between average K_u at 0, 0.25 and 0.5 m depths and average L_v from soil cores of 0.05, 0.1 and 0.15 m below the depths of K_u measurements were 0.63, 0.71 and 0.67 respectively. The higher correlation coefficients from L_v at 0.1 and 0.15 m soil cores result from the fact that the depth of soil wetted during K_u measurements is around 0.1–0.15 m. It is inferred that the K_u at 0.04 m suction and L_v in a moderately swelling duplex soil are positively correlated, as indicated in Fig. 1. Although only three points per curve are included, it should be recalled that each point represents the mean value of 10 pairs of K_u and L_v at a particular depth. The steeper slope of the wheat field relation indicates that a basic difference resulted from cultivation in the soil matrix properties affecting K_u . This may

be due to tillage and crop rather than natural pasture root effects. It is also possible that root and soil shrinkage in the post-harvest wheat field may have created wider intervals between roots and soil which were active in K_u measurements. The evidence of a relationship between K_u and L_v cannot be interpreted as casual; rather, the factors which influence K_u , and those that interact with and are influenced by L_v , are common.

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